

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

### SECTION 1: Identification of the substance/mixture and of the company/undertaking

- **1.1 Product identifier**
- **Trade name:** *Wash Reagent 2*
- **Article number:** 10005487, 4307591
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Pacific Ltd.  
Unit 1101 DCH Commercial Centre  
25 Westlands Road  
Quarry Bay  
Hong Kong
- **Information department:** sales.hongkong@bio-rad.com
- **1.4 Emergency telephone number:** 852-2789-3300

phone: 852-2789-3300 fax: 852-2789-1290

### SECTION 2: Hazards identification

#### · **Emergency overview:**

Colourless, Liquid, Highly flammable liquid and vapour. Harmful in contact with skin. May be harmful if inhaled. Causes skin irritation. Causes serious eye damage. Suspected of causing cancer. Suspected of damaging fertility or the unborn child. May cause drowsiness or dizziness. May cause damage to organs through prolonged or repeated exposure.

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
 Flam. Liq. 2 H225 Highly flammable liquid and vapour.  
 Acute Tox. 4 H312 Harmful in contact with skin.  
 Acute Tox. 5 H333 May be harmful if inhaled.  
 Skin Irrit. 2 H315 Causes skin irritation.  
 Eye Dam. 1 H318 Causes serious eye damage.  
 Carc. 2 H351 Suspected of causing cancer.  
 Repr. 2 H361 Suspected of damaging fertility or the unborn child.  
 STOT SE 3 H336 May cause drowsiness or dizziness.  
 STOT RE 2 H373 May cause damage to organs through prolonged or repeated exposure.

#### · **2.2 Label elements**

- **Labelling according to Regulation (EC) No 1272/2008**  
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms**



GHS02 GHS05 GHS07 GHS08

- **Signal word** *Danger*

(Contd. on page 2)

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

**Trade name: Wash Reagent 2**

(Contd. of page 1)

**Hazard-determining components of labelling:**

trichloromethane

hydrochloric acid

**Hazard statements**

H225 Highly flammable liquid and vapour.

H312 Harmful in contact with skin.

H333 May be harmful if inhaled.

H315 Causes skin irritation.

H318 Causes serious eye damage.

H351 Suspected of causing cancer.

H361 Suspected of damaging fertility or the unborn child.

H336 May cause drowsiness or dizziness.

H373 May cause damage to organs through prolonged or repeated exposure.

**Precautionary statements**

P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER/doctor.

P321 Specific treatment (see on this label).

P362+P364 Take off contaminated clothing and wash it before reuse.

P405 Store locked up.

P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

**2.3 Other hazards****Results of PBT and vPvB assessment****PBT:** Not applicable.**vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

**3.2 Chemical characterisation: Mixtures****Description:** Mixture of the substances listed below with nonhazardous additions.**Dangerous components:**

|                                     |                   |                                                                                                                                                                              |         |
|-------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CAS: 67-64-1<br>EINECS: 200-662-2   | acetone           | Flam. Liq. 2, H225<br>Eye Irrit. 2, H319; STOT SE 3, H336                                                                                                                    | 50-100% |
| CAS: 67-63-0<br>EINECS: 200-661-7   | propan-2-ol       | Flam. Liq. 2, H225<br>Eye Irrit. 2, H319; STOT SE 3, H336<br>Acute Tox. 5, H303; Acute Tox. 5, H333                                                                          | 2.5-5%  |
| CAS: 67-66-3<br>EINECS: 200-663-8   | trichloromethane  | Flam. Liq. 2, H225<br>Acute Tox. 2, H310; Acute Tox. 3, H331<br>Carc. 2, H351; Repr. 2, H361; STOT RE 1, H372<br>Acute Tox. 4, H302; Skin Irrit. 2, H315; Eye Irrit. 2, H319 | 3-<5%   |
| CAS: 7647-01-0<br>EINECS: 231-595-7 | hydrochloric acid | Skin Corr. 1B, H314; Eye Dam. 1, H318<br>STOT SE 3, H335                                                                                                                     | 3-<5%   |
|                                     | Component 7       | Flam. Liq. 3, H226<br>Acute Tox. 4, H302                                                                                                                                     | 2.5-5%  |

(Contd. on page 3)

## Safety data sheet

### according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

**Trade name: Wash Reagent 2**

(Contd. of page 2)

· **Additional information** For the wording of the listed hazard phrases refer to section 16.

#### SECTION 4: First aid measures

##### · 4.1 Description of first aid measures

###### · General information

Immediately remove any clothing soiled by the product.

Symptoms of poisoning may even occur after several hours; therefore medical observation for at least 48 hours after the accident.

· **After inhalation** Supply fresh air; consult doctor in case of complaints.

###### · After skin contact

Generally the product does not irritate the skin.

Immediately wash with water and soap and rinse thoroughly.

Immediately rinse with water.

· **After eye contact** Rinse opened eye for several minutes under running water. Then consult a doctor.

· **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.

· **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.

##### · 4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available.

#### SECTION 5: Firefighting measures

##### · 5.1 Extinguishing media

###### · Suitable extinguishing agents

CO<sub>2</sub>, extinguishing powder or water spray. Fight larger fires with water spray or alcohol resistant foam.

· **For safety reasons unsuitable extinguishing agents** Water with full jet.

##### · 5.2 Special hazards arising from the substance or mixture

During heating or in case of fire poisonous gases are produced.

##### · 5.3 Advice for firefighters

###### · Protective equipment:

Mount respiratory protective device.

Wear self-contained respiratory protective device.

#### SECTION 6: Accidental release measures

##### · 6.1 Personal precautions, protective equipment and emergency procedures

Mount respiratory protective device.

Wear protective equipment. Keep unprotected persons away.

Wear protective clothing.

##### · 6.2 Environmental precautions:

Prevent seepage into sewage system, workpits and cellars.

Do not allow to enter sewers/ surface or ground water.

##### · 6.3 Methods and material for containment and cleaning up:

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

Use neutralising agent.

Dispose contaminated material as waste according to item 13.

(Contd. on page 4)

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

**Trade name: Wash Reagent 2**

(Contd. of page 3)

Ensure adequate ventilation.

**6.4 Reference to other sections**

See Section 7 for information on safe handling

See Section 8 for information on personal protection equipment.

See Section 13 for disposal information.

### SECTION 7: Handling and storage

**7.1 Precautions for safe handling**

Ensure good ventilation/exhaustion at the workplace.

Open and handle receptacle with care.

Prevent formation of aerosols.

**Information about protection against explosions and fires:**

Keep ignition sources away - Do not smoke.

Protect against electrostatic charges.

Keep respiratory protective device available.

**7.2 Conditions for safe storage, including any incompatibilities****Storage****Requirements to be met by storerooms and receptacles:** Store in a cool location.**Information about storage in one common storage facility:** Not required.**Further information about storage conditions:**

Keep receptacle tightly sealed.

Store in cool, dry conditions in well sealed receptacles.

**7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

**Additional information about design of technical systems:** No further data; see item 7.**8.1 Control parameters****Components with limit values that require monitoring at the workplace:****67-64-1 acetone**

|                       |                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------|
| OEL (China (Chinese)) | Short-term value: 450 mg/m <sup>3</sup><br>Long-term value: 300 mg/m <sup>3</sup> |
|-----------------------|-----------------------------------------------------------------------------------|

**67-63-0 propan-2-ol**

|                       |                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------|
| OEL (China (Chinese)) | Short-term value: 700 mg/m <sup>3</sup><br>Long-term value: 350 mg/m <sup>3</sup> |
|-----------------------|-----------------------------------------------------------------------------------|

**67-66-3 trichloromethane**

|                       |                                              |
|-----------------------|----------------------------------------------|
| OEL (China (Chinese)) | Long-term value: 20 mg/m <sup>3</sup><br>G2B |
|-----------------------|----------------------------------------------|

**7647-01-0 hydrochloric acid**

|                       |                                      |
|-----------------------|--------------------------------------|
| OEL (China (Chinese)) | Ceiling limit: 7.5 mg/m <sup>3</sup> |
|-----------------------|--------------------------------------|

(Contd. on page 5)

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

Trade name: Wash Reagent 2

(Contd. of page 4)

### Ingredients with biological limit values:

#### 67-64-1 acetone

|                       |                                                |
|-----------------------|------------------------------------------------|
| BEI (China (Chinese)) | 50 mg/L<br>测试材料: 尿<br>采样时间: 工作班末<br>生物监测指标: 丙酮 |
|-----------------------|------------------------------------------------|

Additional information: The lists that were valid during the creation were used as basis.

### 8.2 Exposure controls

#### Personal protective equipment

#### General protective and hygienic measures

Keep away from foodstuffs, beverages and feed.  
Immediately remove all soiled and contaminated clothing  
Wash hands before breaks and at the end of work.  
Store protective clothing separately.  
Avoid contact with the skin.  
Avoid contact with the eyes.  
Avoid contact with the eyes and skin.

#### Breathing equipment:

In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.  
Not necessary if room is well-ventilated.

#### Protection of hands: Protective gloves.

#### Material of gloves Synthetic gloves

#### Penetration time of glove material

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

#### Eye protection:

Safety glasses  
Tightly sealed goggles.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

#### General Information

#### Appearance:

|                  |                 |
|------------------|-----------------|
| Form:            | Liquid          |
| Colour:          | Colourless      |
| Odour:           | Recognisable    |
| Odour threshold: | Not determined. |

pH-value at 20 °C: 5.5

#### Change in condition

|                                          |              |
|------------------------------------------|--------------|
| Melting point/freezing point:            | undetermined |
| Initial boiling point and boiling range: | 55 °C        |

Flash point: < 0 °C

(Contd. on page 6)

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

Trade name: Wash Reagent 2

(Contd. of page 5)

|                                                  |                                                                                             |
|--------------------------------------------------|---------------------------------------------------------------------------------------------|
| · <b>Flammability (solid, gaseous)</b>           | Not applicable.                                                                             |
| · <b>Ignition temperature:</b>                   | 340 °C                                                                                      |
| · <b>Decomposition temperature:</b>              | Not determined.                                                                             |
| · <b>Self igniting:</b>                          | Product is not selfigniting.                                                                |
| · <b>Explosive properties:</b>                   | Product is not explosive. However, formation of explosive air/vapour mixtures are possible. |
| · <b>Explosion limits:</b>                       |                                                                                             |
| Lower:                                           | 2.6 Vol %                                                                                   |
| Upper:                                           | 13 Vol %                                                                                    |
| · <b>Vapour pressure at 20 °C:</b>               | 233 hPa                                                                                     |
| · <b>Density at 20 °C:</b>                       | 0.82281 g/cm <sup>3</sup>                                                                   |
| · <b>Relative density</b>                        | Not determined.                                                                             |
| · <b>Vapour density</b>                          | Not determined.                                                                             |
| · <b>Evaporation rate</b>                        | Not determined.                                                                             |
| · <b>Solubility in / Miscibility with Water:</b> | Fully miscible                                                                              |
| · <b>Partition coefficient: n-octanol/water:</b> | Not determined.                                                                             |
| · <b>Viscosity:</b>                              |                                                                                             |
| dynamic:                                         | Not determined.                                                                             |
| kinematic:                                       | Not determined.                                                                             |
| · <b>Solvent content:</b>                        |                                                                                             |
| Organic solvents:                                | 90.0 %                                                                                      |
| · <b>9.2 Other information</b>                   | No further relevant information available.                                                  |

### SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**
- **Acute toxicity**
- Harmful in contact with skin.
- May be harmful if inhaled.

(Contd. on page 7)

HK

## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

**Trade name: Wash Reagent 2**

(Contd. of page 6)

· **LD/LC50 values that are relevant for classification:**

**67-64-1 acetone**

|        |      |                    |
|--------|------|--------------------|
| Oral   | LD50 | 5,800 mg/kg (rat)  |
| Dermal | LD50 | 20,000 mg/kg (rbt) |

**67-63-0 propan-2-ol**

|            |          |                       |
|------------|----------|-----------------------|
| Oral       | LD50     | 4,570 mg/kg (rat)     |
| Dermal     | LD50     | 13,400 mg/kg (Rabbit) |
| Inhalative | LC50/4 h | 30 mg/l (rat)         |

**67-66-3 trichloromethane**

|        |      |                 |
|--------|------|-----------------|
| Oral   | LD50 | 908 mg/kg (rat) |
| Dermal | LD50 | 75 mg/kg (rat)  |

**Component 7**

|      |      |                   |
|------|------|-------------------|
| Oral | LD50 | 1,300 mg/kg (rat) |
|------|------|-------------------|

· **Primary irritant effect:**

· **Skin corrosion/irritation**

Causes skin irritation.

· **Serious eye damage/irritation**

Causes serious eye damage.

· **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

· **CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**

· **Germ cell mutagenicity** Based on available data, the classification criteria are not met.

· **Carcinogenicity**

Suspected of causing cancer.

· **Reproductive toxicity**

Suspected of damaging fertility or the unborn child.

· **STOT-single exposure**

May cause drowsiness or dizziness.

· **STOT-repeated exposure**

May cause damage to organs through prolonged or repeated exposure.

· **Aspiration hazard** Based on available data, the classification criteria are not met.

## SECTION 12: Ecological information

· **12.1 Toxicity**

· **Aquatic toxicity:** No further relevant information available.

· **12.2 Persistence and degradability** No further relevant information available.

· **12.3 Bioaccumulative potential** No further relevant information available.

· **12.4 Mobility in soil** No further relevant information available.

· **Additional ecological information:**

· **General notes:**

Water danger class 3 (German Regulation) (Self-assessment): extremely hazardous for water.

Do not allow product to reach ground water, water course or sewage system, even in small quantities.

Must not reach sewage water or drainage ditch undiluted or unneutralised.

Danger to drinking water if even extremely small quantities leak into the ground.

(Contd. on page 8)



## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

Trade name: Wash Reagent 2

(Contd. of page 7)

- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**  
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.

### SECTION 14: Transport information

- |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>· <b>14.1 UN-Number</b></li> <li>· <b>ADR, IMDG, IATA</b></li> </ul>                                                                                                                                                      | <p>UN1993</p>                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>· <b>14.2 UN proper shipping name</b></li> <li>· <b>ADR</b></li> <li>· <b>IMDG, IATA</b></li> </ul>                                                                                                                       | <p>1993 FLAMMABLE LIQUID, N.O.S. (vapour pressure at 50°C not more than 110 kPa) (ACETONE, CHLOROFORM)<br/>1993 FLAMMABLE LIQUID, N.O.S. (vapor pressure at 50°C at most 110 kPa) (ACETONE, CHLOROFORM)<br/>FLAMMABLE LIQUID, N.O.S. (ACETONE, CHLOROFORM)</p> |
| <ul style="list-style-type: none"> <li>· <b>14.3 Transport hazard class(es)</b></li> <li>· <b>ADR, IMDG, IATA</b></li> <li>· <b>Class</b></li> <li>· <b>Label</b></li> </ul>                                                                                     | <p>3 Flammable liquids.<br/>3</p>                                                                                                                                                                                                                              |
| <ul style="list-style-type: none"> <li>· <b>14.4 Packing group</b></li> <li>· <b>ADR, IMDG, IATA</b></li> </ul>                                                                                                                                                  | <p>II</p>                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>· <b>14.5 Environmental hazards:</b></li> <li>· <b>Marine pollutant:</b></li> </ul>                                                                                                                                       | <p>No</p>                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>· <b>14.6 Special precautions for user</b></li> <li>· <b>Hazard identification number (Kemler code):</b></li> <li>· <b>EMS Number:</b></li> <li>· <b>Segregation groups</b></li> <li>· <b>Stowage Category</b></li> </ul> | <p>Warning: Flammable liquids.<br/>33<br/>F-E, S-E<br/>Liquid halogenated hydrocarbons<br/>B</p>                                                                                                                                                               |
| <ul style="list-style-type: none"> <li>· <b>14.7 Transport in bulk according to Annex II of Marpol and the IBC Code</b></li> </ul>                                                                                                                               | <p>Not applicable.</p>                                                                                                                                                                                                                                         |

(Contd. on page 9)



## Safety data sheet

according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

Trade name: Wash Reagent 2

(Contd. of page 8)

### · Transport/Additional information:

#### · ADR

#### · Limited quantities (LQ)

1L

#### · Excepted quantities (EQ)

Code: E2

Maximum net quantity per inner packaging: 30 ml

Maximum net quantity per outer packaging: 500 ml

#### · Transport category

2

#### · Tunnel restriction code

D/E

#### · IMDG

#### · Limited quantities (LQ)

1L

#### · Excepted quantities (EQ)

Code: E2

Maximum net quantity per inner packaging: 30 ml

Maximum net quantity per outer packaging: 500 ml

#### · UN "Model Regulation":

UN 1993 FLAMMABLE LIQUID, N.O.S. (VAPOUR PRESSURE AT 50°C NOT MORE THAN 110 KPA) (ACETONE, CHLOROFORM), 3, II

## SECTION 15: Regulatory information

### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

#### · Directive 2012/18/EU

#### · Named dangerous substances - ANNEX I None of the ingredients is listed.

#### · Seveso category P5c FLAMMABLE LIQUIDS

#### · Qualifying quantity (tonnes) for the application of lower-tier requirements 5,000 t

#### · Qualifying quantity (tonnes) for the application of upper-tier requirements 50,000 t

#### · National regulations

#### · Technical instructions (air):

| Class | Share in % |
|-------|------------|
| I     | 2.5-5      |
| III   | 2.5-5      |
| NK    | 50-100     |

#### · Water hazard class: Water danger class 3 (Self-assessment): extremely hazardous for water.

#### · 15.2 Chemical safety assessment: A Chemical Safety Assessment has not been carried out.

## SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

#### · Department issuing SDS: Environmental Health and Safety.

#### · Contact:

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

(Contd. on page 10)

## Safety data sheet

### according to 1907/2006/EC, Article 31

Printing date 15.01.2021

Revision: 31.07.2020

**Trade name: Wash Reagent 2**

(Contd. of page 9)

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

· **Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Flam. Liq. 2: Flammable liquids – Category 2

Flam. Liq. 3: Flammable liquids – Category 3

Acute Tox. 4: Acute toxicity - oral – Category 4

Acute Tox. 5: Acute toxicity - oral – Category 5

Acute Tox. 2: Acute toxicity - dermal – Category 2

Acute Tox. 3: Acute toxicity - inhalation – Category 3

Skin Corr. 1B: Skin corrosion/irritation – Category 1B

Skin Irrit. 2: Skin corrosion/irritation – Category 2

Eye Dam. 1: Serious eye damage/eye irritation – Category 1

Eye Irrit. 2: Serious eye damage/eye irritation – Category 2

Carc. 2: Carcinogenicity – Category 2

Repr. 2: Reproductive toxicity – Category 2

STOT SE 3: Specific target organ toxicity (single exposure) – Category 3

STOT RE 1: Specific target organ toxicity (repeated exposure) – Category 1

STOT RE 2: Specific target organ toxicity (repeated exposure) – Category 2

· **\* Data compared to the previous version altered.**